

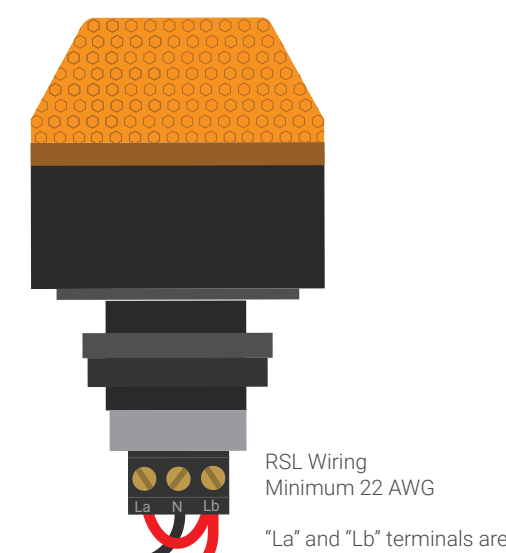
The FRC Control System - A Primer

Lays out and explains the essential control system hardware that is needed to make an FRC robot run

Please consult the Regular FRC Control System Diagrams for a more complete implementation of the system

Robot Signal Light

Allows a robot to visually communicate its status to the field and your team



Ethernet connection to the Systemcore
Connection uses the "Robot" or "RIO" Port

Powering your Radio

Power over Ethernet (PoE)-only setup shown
Please consult the Game Manual for more information

WCP-2142 shown

Radio

V2 radio example shown
Enables wireless communication from your robot to the field, and to your team's Driver's Station

Robot Communication

Main Power

High Current Channels

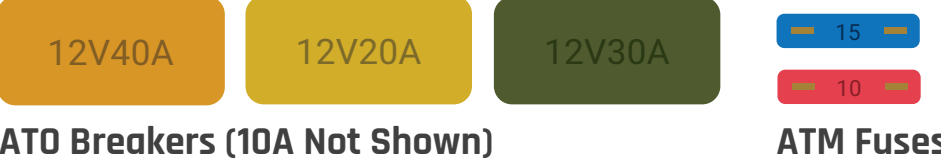
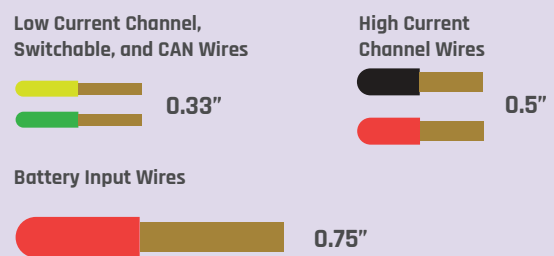
Up to 40A continuous load. Used to power motors, pneumatics controllers, and everything else

Low Current Channels

Up to 15A continuous, 20A peak. If available, it is usually used to power the systemcore, radio and other low current devices

PDH Wire Strip Guide

Following these strip lengths for the Power Distribution Hub wire connections ensures that your connections are robust and reliable. Illustrated Lengths are 1:1 scale



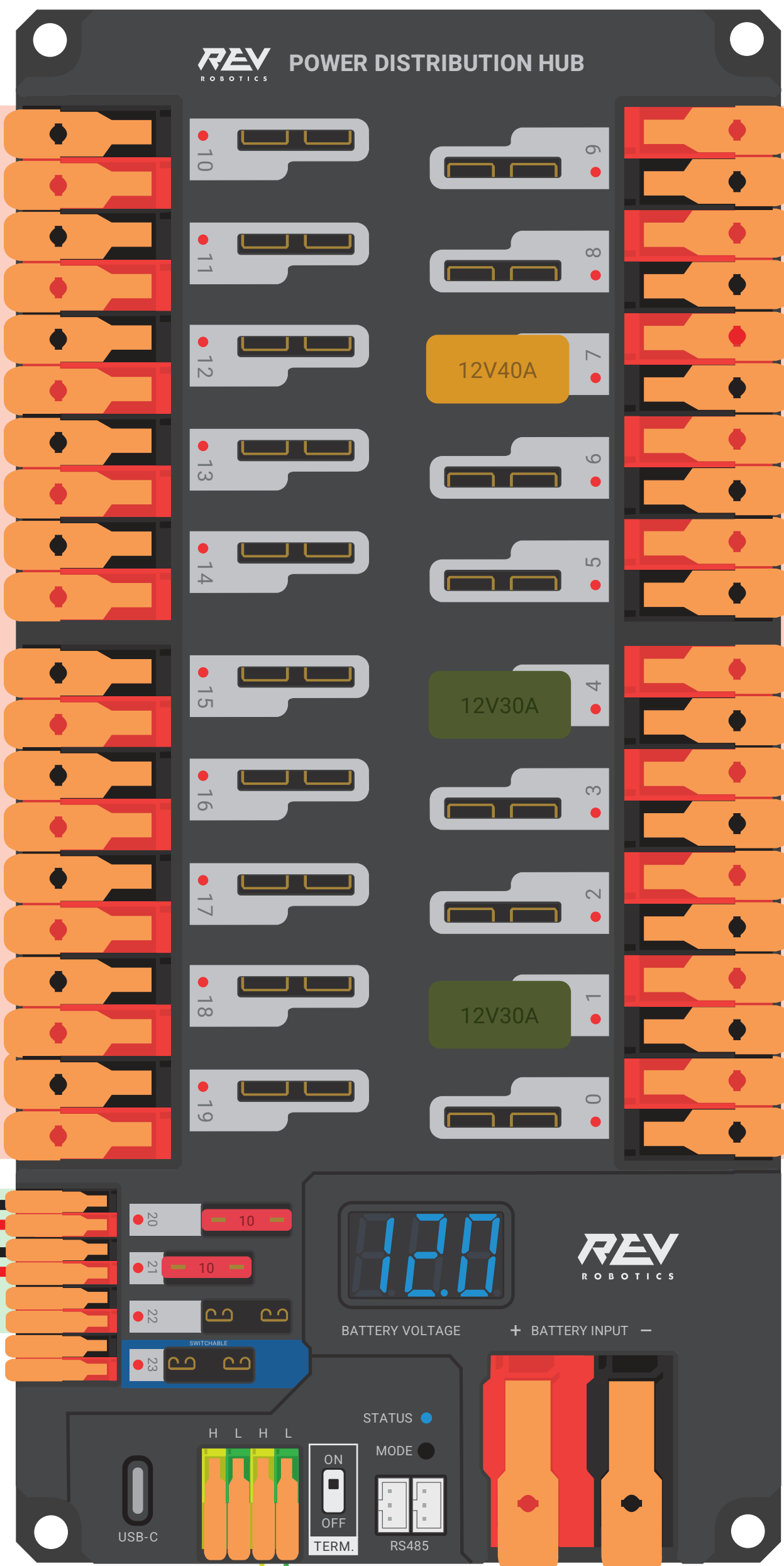
Circuit Breakers / Fuses

Comes in ATO (High Current) or ATM (Low Current) form factors. These slot in to supply power as well as give current protection to the channel its connected to

Power Distribution Board

Example Shown is a REV Power Distribution Hub

Distributes battery power into the various components of your robot



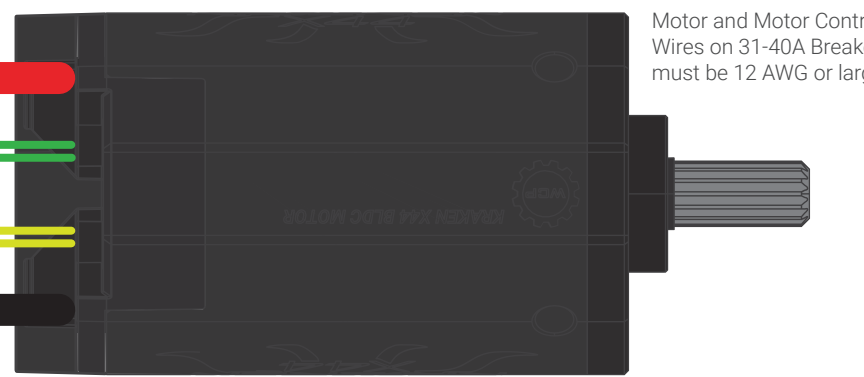
Motor Controllers

Controls the motors that your robot has. They come in different types, and can be controlled either through the CAN Bus or through PWM

Integrated Controllers

Example shown is a Talon FX on a Kraken X44 Motor

These controllers are usually integrated to a specific motor

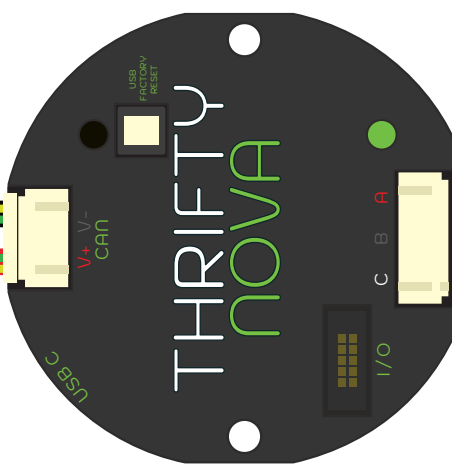


Motor and Motor Controller Wires on 31-40A Breakers must be 12 AWG or larger

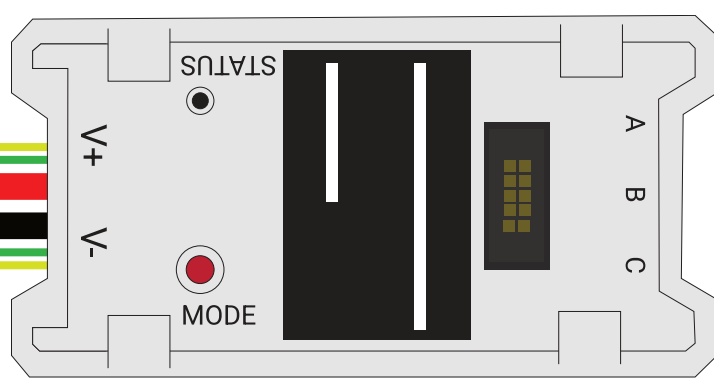
Discrete Controllers

Nova and Spark Max examples shown

These controllers can be paired with any compatible motor



Motor and Motor Controller Wires on 21-30A Breakers must be 14 AWG or larger



Motor Control

The Battery

A sealed SLA-type battery, it stores and provides the power needed for a robot to run for a Full Match (2:30 Minutes). **Replaced with a recharged one every match.**

120A Main Breaker

Protects the robot from drawing too much current at the battery level. This is also your robot's power switch



Battery lugs must be crimped and properly insulated

Systemcore

The robot's brain. It runs your team's code, telling your robot how to move, respond, and perform defined tasks



Ensuring Reliability

Controlling Devices

At a basic level, the systemcore can issue commands into a device using two ways

CAN Bus
Uses a standard messaging protocol to communicate between connected devices in the robot

Pulse Width Modulation
Controls power given to a device by using rapid and variable pulses of power

Sensors

Making your robot aware

Sensors detect movements or certain changes in your robot or its surroundings, allowing your robot to react and perform tasks accordingly

Read more on the different types of sensors and how you could utilize them at: <https://docs.wpilib.org>

Limit Switch
Generic Shown

Analog Encoder
TTB-0040 Shown

USB Camera
TTB-0370 Shown

BASIC FRC CONTROL SYSTEM

V.4.000PR.TBT

TEAM 3161

Printed and Distributed by Thriftybot

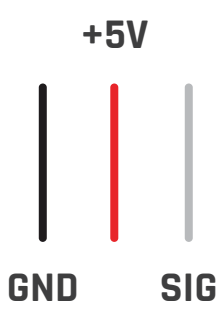
Power

12V DC Main



PWM

26 AWG Minimum



CAN

28 AWG Minimum



Wires

American Wire Gauge (AWG)

Minimum Gauge per connection type shown.



Always practice proper safety precautions and practices when working with electrical systems.

More Information about the FRC Control System can be found at: <https://docs.wpilib.org>

KEEP IN MIND